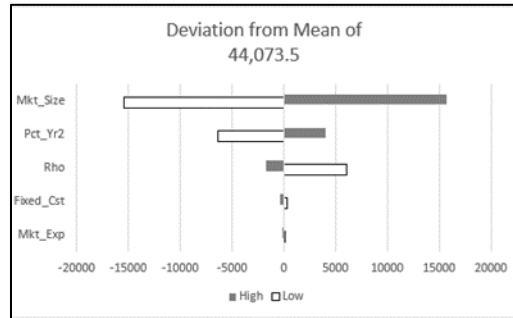


Sensitivity Analysis for ChanceCalc

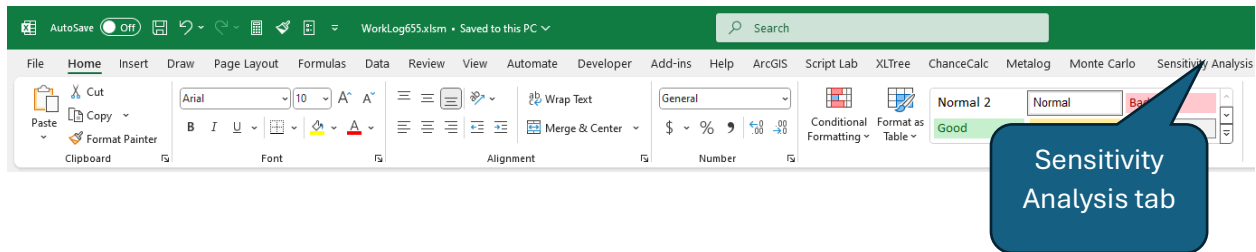


By Dave Empey

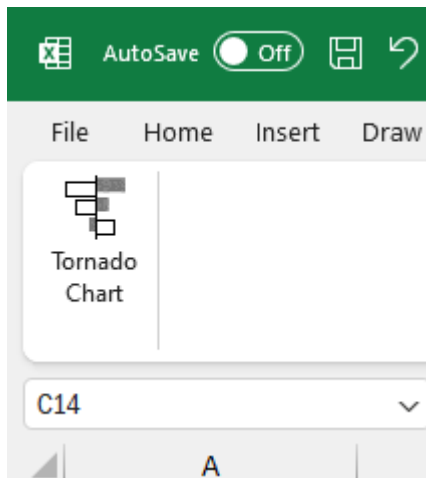
9-24-2024

NOTE: If you have a ChanceCalc or SIPmath model with a PMTable set up, this will overwrite your existing PMTable!

The Sensitivity Analysis tool is found under the tab named Sensitivity Analysis on the Excel Ribbon.



Click the tab to show the Tornado Chart sensitivity analysis button.



Press the Tornado Chart button to open the Tornado Chart parameter entry form.

Select Parameters

Output

Location:

Output Name:

Low %:

Chart Location

Inputs

Uniform Location:

Variable Name:

Selected Inputs:

Select Input

Remove Input

OK Cancel

Note: if you select a cell with a formula before pressing the button, that cell address will be filled in as the Output cell Location.

Click in the Output Location field, and then select the cell containing the formula whose sensitivity you want to analyze. The Output Location field will be filled in with the cell address. In the example below, the Output Location cell is 'Control Panel'!\$B\$28.

The screenshot shows the Sensitivity Analysis tool interface. The background is a spreadsheet titled 'Product Launch Decision' with various tables for parameters, decisions, and objectives. The 'Control Panel' tab is active, showing a table with the following data:

Parameter	Units	In Use	Index	Worst	Base	Best
Mkt_Size	k	1.8913099	1	2	3	0.3972
Mkt_Exp	k	54.071994	70	50	30	0.6829
Fixed_Cst	k	216.19678	300	200	100	0.6489
Rho	NA	80.174622	1000	20	10	0.1279

The 'Control Panel' table shows the following data:

Unit Price	Max NPV (\$ k)
20	\$47,441
25	\$47,441
30	\$47,441
35	\$47,441
40	\$47,441
45	\$47,441

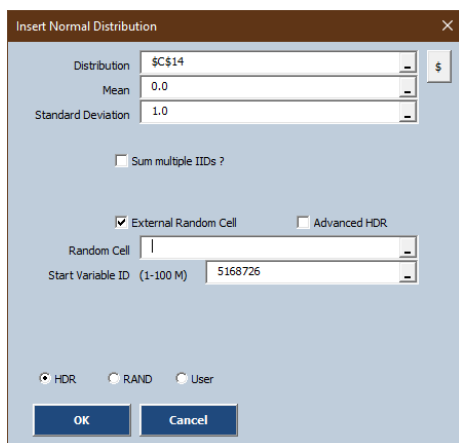
The 'Sensitivity Analysis' dialog box is open, showing the 'Output' field with the location 'Control Panel!\$B\$28' and the 'Output Name' field. The 'Inputs' field is empty. The 'Chart Location' field is also empty. The 'Low %' field is set to 5%.

Click in the Output Name field. You can either click on a cell containing the Output Name, or type in the Output Name you want. The name should consist of letters, numbers, and underscore characters and have no spaces.

In Tornado Chart sensitivity analysis, you vary a set of Inputs by setting each one successively to a low value, perhaps the 5th percentile of the input range, and then to a high value, which for this tool will be 100%-the low value %.

Click the Low% field and then click the cell with the low % value you want to use for your sensitivity analysis.

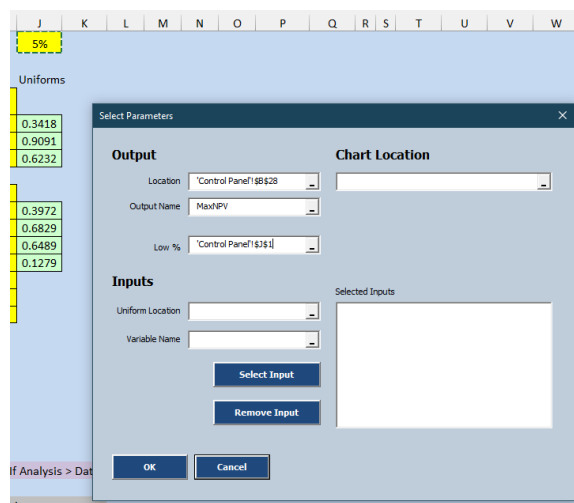
Input variables:



The 'Insert Normal Distribution' dialog box is shown. It has fields for 'Distribution' (set to '\$C\$14'), 'Mean' (0.0), and 'Standard Deviation' (1.0). There is a checkbox for 'Sum multiple IIDs?' which is unchecked. Below that, there is a checkbox for 'External Random Cell' which is checked, and an unchecked checkbox for 'Advanced HDR'. The 'Random Cell' field is empty. The 'Start Variable ID (1-100 M)' field is set to '5168726'. At the bottom, there are three radio buttons: 'HDR' (selected), 'RAND', and 'User'. 'OK' and 'Cancel' buttons are at the bottom.

With this tool, Input Variables must be determined by External Uniform random variables, such as are created by the SIPmath Generate Input menu option; for example the Normal Distribution.

That is, you must choose the External Random Cell checkbox, as shown in the example here.



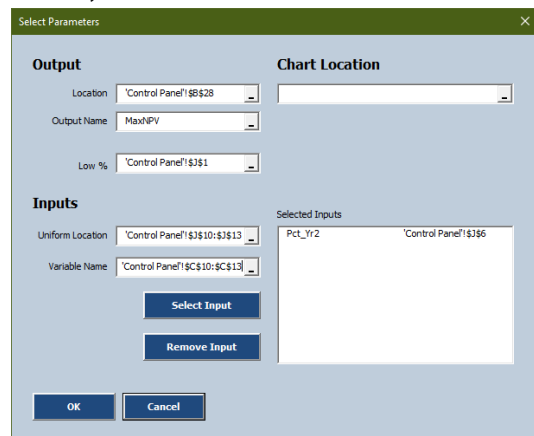
The 'Select Parameters' dialog box is shown. It has two main sections: 'Output' and 'Chart Location'. The 'Output' section has fields for 'Location' (set to 'Control Panel!\$B\$28'), 'Output Name' (set to 'MaxNPV'), and 'Low %' (set to 'Control Panel!\$J\$11'). The 'Chart Location' section is empty. Below these are the 'Inputs' section, which has fields for 'Uniform Location' and 'Variable Name', both empty. To the right of these fields is a 'Selected Inputs' list, which is currently empty. At the bottom are 'Select Input' and 'Remove Input' buttons, and 'OK' and 'Cancel' buttons at the very bottom.

Selecting Input Variables:

Since typically you will want to determine the sensitivity to several different input variables, the Probability Management Tornado Chart tool allows you to create a list of several variables to analyze.

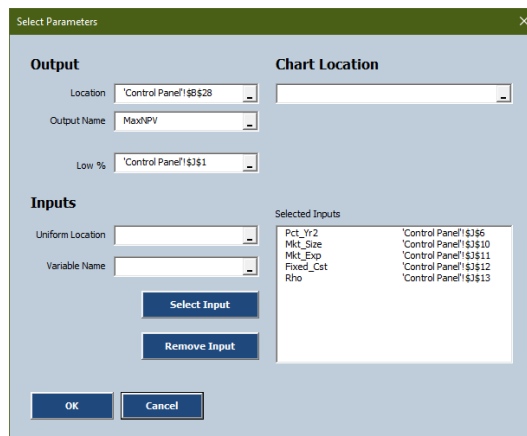
To add Input Variables to the list, select the Uniform Location (the External Uniform cell mentioned above), the Variable Name, and press the Select Input button on the form.

If you have a range of input variables, you can add them all at once by selecting them (and their names) and pressing the Select Input button. Here the user has already added an input variable at cell J6, and is about to add 4 more variables at cells J10:J13.



The 'Select Parameters' dialog box is shown. The 'Output' and 'Chart Location' sections are the same as in the previous image. In the 'Inputs' section, the 'Uniform Location' field is now set to 'Control Panel!\$J\$10:\$J\$13' and the 'Variable Name' field is set to 'Control Panel!\$C\$10:\$C\$13'. The 'Selected Inputs' list now contains two entries: 'Pct_Yr2' and 'Control Panel!\$J\$6'. The 'Select Input' and 'Remove Input' buttons are still present.

The cells have been added.

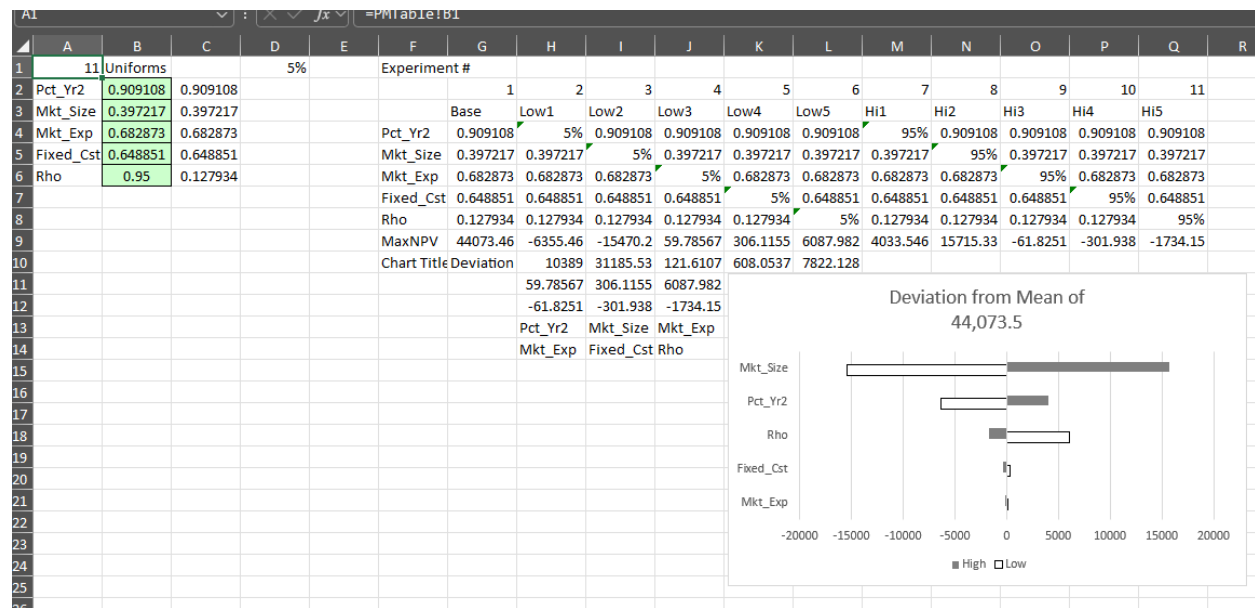


The 'Select Parameters' dialog box is shown. The 'Output' and 'Chart Location' sections are the same. In the 'Inputs' section, the 'Uniform Location' and 'Variable Name' fields are empty. The 'Selected Inputs' list now contains five entries: 'Pct_Yr2', 'Mkt_Size', 'Mkt_Exp', 'Fixed_Cst', and 'Rho'. The 'Select Input' and 'Remove Input' buttons are still present.

If you make a mistake adding input cells, you can remove the mistaken entry by clicking the mistaken entry in the list of selected inputs and pressing the Remove Input button.

You can specify a location for the tornado chart in the Chart Location cell; if you leave this blank, the chart will be created on a new page called PM_Sensitivity created by the tool.

When you have selected all your sensitivity parameters, press the OK button and the tool will create a tornado chart based on the model and the cell and % values you entered.



This chart is live, in that it will update when parameters are changed within the model.